Amazing DNA – reading the DNA code 4a

1 Use the word bank to complete the information about DNA.

Word bank

- differs
- code
- important
- together
- strand
- helix
- significant
- discovery
- nucleotides
- things
- zygote
- organic
- egg

Your life began when one _____ and one sperm united to form a _____ – from a single cell to you! How did this single cell turn into thousands and thousands of different cells with specialised functions? Where did the instructions for this come from?

The answer lies with DNA. DNA is short for deoxyribonucleic acid. It is an _____ molecule that contains all the genetic information needed to make you, and all other living things. The DNA of all living _____ is unique.

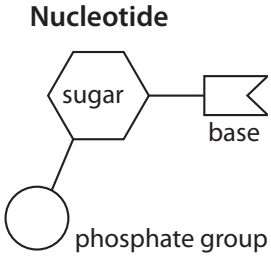


Figure 1a

The _____ of DNA was led by James Watson and Francis Crick in 1953, with the help of Rosalind Franklin and Maurice Wilkins. It was one of the most famous and _____ discoveries in Science.

The basic unit of DNA is the nucleotide. (See Fig 1a.) DNA is made up of millions of _____ linked together like a chain. A nucleotide consists of a sugar, a phosphate and a nitrogenous base.

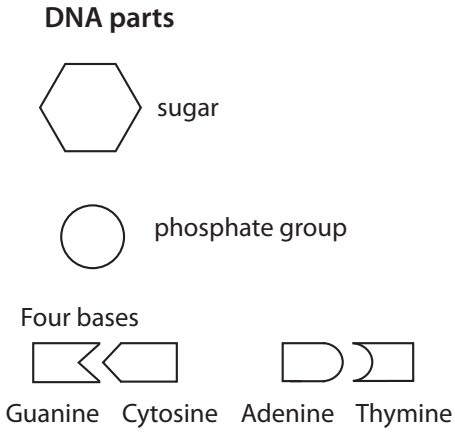


Figure 1b

There are four different nitrogenous bases in DNA: adenine (A), thymine (T), cytosine (C) and guanine (G). The bases pair up, always **A – T** and **G – C**. (See Fig 1b.) The order of these bases is what determines the chemical code. One _____ of nucleotides is joined together to another strand of nucleotides with the help of the bases. The two strands of nucleotides spiral with each other to make a shape called the _____. DNA is often referred to as a double helix because of the two strands of nucleotides (see Fig 2). There is another molecule like DNA in every cell. It is called RNA (ribonucleic acid). RNA is a single strand and contains the sugar ribose, whereas DNA has deoxyribose. Another difference is that RNA does not have the base thymine; instead it has uracil, symbol U.

The order of the bases in DNA is very _____, as this codes for protein. Proteins are long chains of amino acids joined _____ like beads on a chain. In 1960, 20 different amino acids were identified. It was discovered that three bases of m-RNA code for one amino acid (see Table 1).

Two strands of nucleotides: DNA

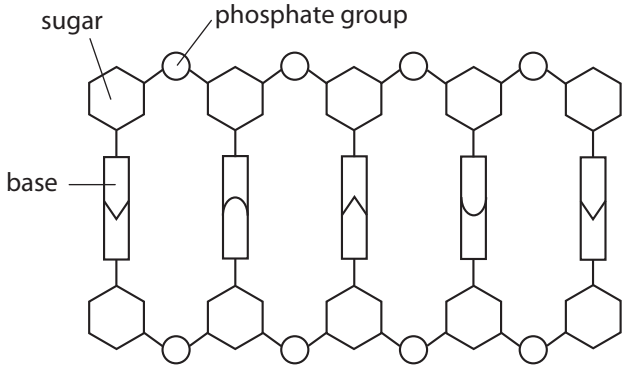


Figure 2

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Amazing DNA – reading the DNA code

4b

All the codes were organised into a table (see Table 1) which shows the three bases (each abbreviated by a capital letter) in front of an amino acid (abbreviated to three letters). The full names of the amino acids are listed in Table 2; 'stop' indicates the end of an amino acid sequence – the protein is complete.

Remember, DNA is **opposite** to the _____ of m-RNA. That is, m-RNA is the complementary copy of DNA. But it _____ in one base: instead of **thymine**, RNA has **uracil**.

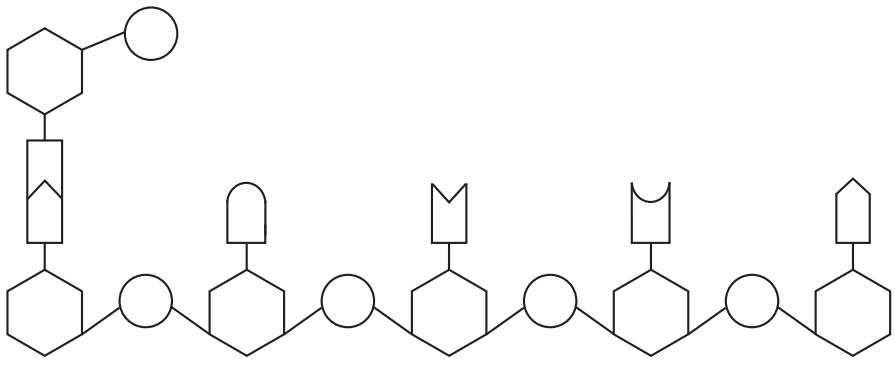
Table 1

m-RNA amino acid triplet codes			
UUU Phe	UCU Ser	UAU Tyr	UGU Cys
UUC Phe	UCC Ser	UAC Tyr	UGC Cys
UUA Leu	UCA Ser	UAA stop	UGA stop
UUG Leu	UCG Ser	UAG stop	UGG Try
CUU Leu	CCU Pro	CAU His	CGU Arg
CUC Leu	CCC Pro	CAC His	CGC Arg
CUA Leu	CCA Pro	CAA Gln	CGA Arg
CUG Leu	CCG Pro	CAG Gln	CGG Arg
AUU Ile	ACU Thr	AAU Asn	AGU Ser
AUC Ile	ACC Thr	AAC Asn	AGC Ser
AUA Ile	ACA Thr	AAA Lys	AGA Arg
AUG Met	ACG Thr	AAG Lys	AGG Arg
GUU Val	GCU Ala	GAU Asp	GGU Gly
GUC Val	GCC Ala	GAC Asp	GGC Gly
GUA Val	CGA Ala	GAA Glu	GGA Gly
GUG Val	GCG Ala	GAG Glu	GGG Gly

Table 2

Amino acid names	
Ala = alanine	Leu = leucine
Arg = arginine	Lys = lysine
Asn = asparagine	Met = methionine
Asp = aspartic acid	Phe = phenylalanine
Cys = cysteine	Pro = proline
Gln = glutamine	Ser = serine
Glu = glutamic acid	Thr = threonine
Gly = glycine	Trp = tryptophan
His = histidine	Tyr = tyrosine
Ile = isoleucine	Val = valine

2 Draw the complementary strand on the diagram below, then label each part of the nucleotide chain. The first nucleotide has been drawn for you.



3 List three differences between DNA and RNA.

_____

_____

4 What is a *nucleotide*?_____

5 What do you think would happen if m-RNA copied the DNA code incorrectly?

_____

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6 Use Figure 2 to work out all the different codes for each amino acid. The first row has been done for you.

Amino acid	Complete name	Possible codes
Val	Valine	GUU, GUA, GUG, GUC
Trp	Tryptophan	
Thr	Threonine	
Ser		
Pro		
Phe		
Met		
Leu		
Ile		
Glu		
Cys		
Asp		

7 The following amino acids could produce proteins for eye colour.

Lys-Val-Tyr-Ser-Pro-Leu-Ile → blue eyes

Lys-Val-Tyr-Pro-Ser-Leu-Ile → hazel eyes

Lys-Pro-Tyr-Pro-Phe-Ser-Ile → light brown eyes

Lys-Pro-Tyr-Val-Phe-Ser-Ile → dark brown eyes

Use Table 1 to help you decode the following DNA or RNA sequences to reveal the eye colour of each person. *Hint:* Their DNA sequence is their eye colour gene.

	Person 1	Person 2
DNA code	TTC GGG ATG CAC AAA AGG TAG	TTC CAT ATG GGC AGA AAC TAA
m-RNA complementary code	____ ____ ____ ____ ____ ____ ____	____ ____ ____ ____ ____ ____ ____
Protein		
Eye colour		

	Person 3	Person 4
DNA code	____ ____ ____ ____ ____ ____ ____	____ ____ ____ ____ ____ ____ ____
m-RNA complementary code	AAG CCU UAC CCG UUU UCC AUC	AAG GUA UAC UCU CCG UUG AUU
Protein		
Eye colour		

8 Sometimes changes happen to DNA. For example, the DNA code of Person 2 – Suzan – TTC CAT ATG GGC AGA AAC TAA has changed to the following code due to a flip in the DNA bases: TTC CAT ATG AGA GGC AAC TAA. This is called a mutation.

a Find the complementary m-RNA code of Suzan's new DNA code._____

b Work out the protein chain._____

c What colour are Suzan's eyes now?_____

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